

Working and travelling with teams

with LIAM WEST

Talent wins games, but teamwork and intelligence win championships.

Michael Jordan

Within sports medicine, involvement in team care is a very rewarding yet challenging experience. Working alongside other clinicians and disciplines enables individuals to build upon their own skills and translate this into their everyday practice. Working with athletes on a regular basis helps the clinician to understand the psychological pressures placed upon them and appreciate the physical demands of the particular sport.

Clinicians working in a team environment must adapt and fit into the 'team chemistry'. This involves adhering to team rules and customs while appreciating that the clinician is a small cog in a very large wheel, where the players and the coach are the key people.¹ To be accepted by the coaches, athletes and other members of the medical team, the clinician needs to be in regular attendance during training sessions and competition. This in return offers the relatively unique opportunity to closely observe and understand mechanisms of injury before monitoring the progress of these injuries during rehabilitation.

Travelling with a team presents the sports medicine clinician with a considerable challenge.² In order to provide quality medical support on the road, clinicians must be flexible, innovative and fulfil a number of roles, as not all members of the medical team will be able to travel and less equipment will be available. Travelling with a team often involves working long hours in less than ideal conditions with sportspeople and coaches who are under great stress due to the demands of competition and travel.³ Nonetheless, travelling with a team is fun, a great opportunity to learn, and offers the chance to broaden your horizons by seeing the world.

THE MEDICAL SUPPORT TEAM

The size and make-up of the medical support team often depends on the size of the sporting team, the competition standard and financial considerations. Frequently, the support team will consist of just one individual, who may be a physiotherapist, physician, massage therapist or trainer. Specialists from various branches of medicine can contribute to the sports and exercise medicine (SEM) team.⁴⁻⁶ It is imperative that a solo clinician develops a network of colleagues who can assist where additional specialised management is indicated. Not all members of the medical support team will be remunerated and the ethical issues facing professional teams' clinicians are slightly different from those of volunteer clinicians.⁷⁻¹⁰

KEY ATTRIBUTES OF A SUCCESSFUL MEDICAL TEAM

In order to add value to athletes, the medical team often consists of representatives from different health disciplines. In an ideal scenario, the professional sporting team should have access to the services of an SEM physician, physiotherapist, massage therapist, podiatrist, dietitian, psychologist, orthopaedic surgeon, cardiologist and sports trainer as well as the coaching, sports science and fitness staff. Whoever is responsible for assembling such a support team must ensure that all individuals have high professional standards and work well collectively.¹¹ This leader does not necessarily need to be the team physician; often the physiotherapist or sports scientist is better placed to lead the medical team as they work full time with the sporting team. The appointed medical team lead should take ultimate responsibility for difficult management decisions and the smooth running of the group.

The best medical teams have clearly defined roles and duties that help to avoid conflict. The clinician has a primary responsibility to the athlete, but must determine and define his or her responsibilities to the coach, the team management and the fellow support staff before taking up the position. Medical teams should actively encourage the 'cross pollination' of ideas from different medical disciplines. Effective communication with each other, the athletes and the coaching staff is the cornerstone of success for a medical team.

MEDICAL INDEMNITY AND TRAUMA TRAINING

We live in a world of litigation, and the sports medicine arena is not exempt. It is imperative that sports clinicians arrange medical indemnity to cover their entire scope of practice. Without this cover, the financial and reputational risks can be devastating. It is the responsibility of the team clinician to ensure that his or her medical indemnity is not only current, but also indemnifies them in any state, county or country to which team competition may take them. In addition to indemnity, it is important that medical team members (not just the physician) undertake an appropriate advanced resuscitation and emergency aid course for their sport so that they can competently care for a seriously injured or ill athlete. Without adequate training, indemnity may be null and void.

WHERE DOES THE MEDICAL TEAM WORK?

The medical team is required to operate in various settings including at the training ground, at competition venues, and during travel. At each location the team must ensure that it has adequate equipment for each member to perform his or her role—see below for information on the pitchside medical bag. The club medical room should ideally be a clean room that has a door that can be locked to enable privacy and confidentiality. It should have at least one examination couch, facilities to enable the clinician to clean his or her hands at a sink or with alcohol hand gel, and all the medical equipment required for the particular sport.

The team clinician must ensure that medications and other medical equipment, such as oxygen, are stored safely and securely. If the clinician is expected to perform procedures such as suturing, he or she should aim to have an area to do so under sterile conditions while being able to safely dispose of sharps and blood-contaminated products. Clinicians should consider decorating the medical room with anatomy posters to help explain injuries, and evidence-based infographics to educate athletes about various topics within sports medicine such as the importance of sleep or adequate hydration.

MEDICAL EQUIPMENT

The contents of the medical bag will vary depending on the type of sport, access to other facilities, and the clinician's own preferences. It is often impossible to carry all required items in one bag and therefore equipment should be organised into storage within the medical room, a pitchside suitcase, a medical bag and a hip bag that contains the essentials to provide athletes with symptom relief or immediate care until a more comprehensive kit can be accessed.¹² An exhaustive list of contents of medical equipment for a contact sports team without immediate access to more sophisticated facilities can be found in the box below. When travelling with a team, the contents of the medical kit will vary depending on the make-up of the medical support team, the size of the overall team and the local facilities available.

TEAM CARE THROUGHOUT THE SEASON

The team clinician must tailor athlete care according to the stage of competition – pre-competition, during competition and post-competition. In the pre-competition phase there should be a full assessment of all team members based around a comprehensive history and examination (Chapter 46). This screening should evaluate for medical illness and musculoskeletal conditions that may need further follow-up. This screening will often be carried out in conjunction with a fitness assessment. A similar assessment should be performed on new team recruits and also on the entire team at the end of a season to plan appropriate individual off-season treatment and rehabilitation programs. Arrangements for how this will be monitored by the sports clinician should also be made. During the season, post-game/competition injury clinics help to diagnose and plan treatment of acute injuries and illnesses. Where possible, these clinics should be attended by both a team physician and physiotherapy staff.

Core principles providing care for a team

The medical team should look to develop various core 'team principles' that underpin good sports medicine care. An athlete's right to confidentiality must be strictly adhered to by ensuring that their medical problems are not discussed with other team members, officials or the media without the player's permission.¹³ Good record keeping is important for patient care and medico-legal purposes. There are many excellent software programs that allow practitioners to maintain records on a laptop computer, which is particularly useful when travelling with a team. The sports clinician should consider keeping two sets of password-encrypted notes: one for musculoskeletal injuries that can be seen by other members of the treating medical team, and the other for medical and mental health issues to which only

Contents of the medical bag

Diagnostic instruments

Oral/rectal thermometer
 Stethoscope
 Blood pressure cuff
 Ophthalmoscope
 Otoscope (auroscope) & ear pieces
 Pencil torch
 Tongue depressors
 Oxygen saturation (SATS) probe
 Tendon hammer
 Portable ultrasonography device
 Snellen eye chart & Ishihara plates
 SCAT3™ forms

Medications

Oral analgesics (e.g. paracetamol [acetaminophen], aspirin, codeine, tramadol)
 Intravenous analgesics (morphine, tramadol)
 Adrenaline (epinephrine) for anaphylaxis (& hydrocortisone)
 NSAIDs (nonsteroidal anti-inflammatory drugs)
 Antibiotics (e.g. amoxycillin, erythromycin, flucloxacillin, doxycycline, metronidazole)
 Antacid tablets
 Anti-nausea tablets (e.g. prochlorperazine [oral/IM], metoclopramide)
 Antidiarrhoeal agent (e.g. loperamide)
 Faecal softeners
 Antihistamines
 Bronchodilators (e.g. salbutamol inhaler, beclomethasone inhaler)
 50% glucose solution
 Sedatives and hypnotics
 Nasal spray decongestants
 Eye and ear antibiotic solutions
 Ear wax softener
 Throat lozenges
 Cough mixture
 Creams/ointments: antifungal, antibiotic, corticosteroid, anti-inflammatory
 Tetanus toxoid
 GTN sublingual spray
 Corticosteroid injections

Sutures/dressings

Needle holders

Forceps

Scissors: nail clippers, small sharp scissors, tape scissors
 Syringes (various sizes)
 Needles (various sizes)
 Sutures (various sizes), absorbable & non-absorbable
 Suture cutters
 Scalpels
 Local anaesthetics: lignocaine (lidocaine) with and without adrenaline (epinephrine)
 Steri-strips™ (various sizes)
 Alcohol swabs
 Gauze swabs
 Dressings packs
 Antiseptic solution
 Tincture of benzoin
 Dressing pads (various sizes)
 Low-adherent dressing (e.g. Jelonet™)
 Band-Aid® plastic strips
 Crepe bandages
 Tube gauze (Tubigrip™)
 Sterile gloves, goggles & mask

Resuscitation and emergency equipment

Defibrillator
 Resuscitation medications (e.g. adrenaline, amiodarone)
 Oropharyngeal & nasopharyngeal airways (various sizes)
 Bag valve mask (BVM)
 Oxygen with suitable delivery masks
 Emergency cricothyroid set
 Portable suction device
 Cannulation equipment
 Intravenous (IV) sodium chloride
 Intravenous (IV) giving set
 Rectal diazepam (for seizures)
 EpiPen®

Blood management

Gauze swabs
 Klatostat® for lacerations
 Klatostat strings & Rapid Rhino™ for epistaxis
 Blood tape (Coban™)
 Co-phenylamine/tranexamic acid nasal spray
 Betadine® spray
 Vaseline®

Other equipment

Bolt cutters/screwdriver
 Air/box splints
 Triangular bandages (sling)
 Cotton-tipped applicators
 Rigid sports tape (various sizes)
 Elastic adhesive tape (various sizes)
 Compression bandage (various sizes)
 Adhesive foam pad
 SAM splint
 Blister pads
 Coolant spray
 Crutches
 Deep Heat® muscle warming spray
 Finger splints
 Adjustable cervical collar (soft & hard)
 Spinal board with straps
 Eye kit: irrigation solution, fluorescein, eye patches, local anaesthetic/dilating eye drops, contact lens container
 Sunscreen
 Bonjela mouth ulcer cream
 Massage oil/heat rubs
 Electrotherapy (e.g. TENS, portable laser)
 Portable couch
 Alarm clock
 Plaster of Paris/thermoplastic casting equipment

Miscellaneous equipment

Pen and paper
 Urine reagent strips and urine sample pots
 Safety pins
 Tampons
 Condoms
 Contaminated needle container
 Acupuncture needles
 Foam rollers & rehabilitation equipment
 Spare shoelaces
 Spare mouth guard
 Flexible orthoses
 Batteries
 Safety razor
 Plastic bags & cling film for ice
 Transformer & dual voltage connector (if appropriate)
 Glucose sweets (for hypoglycaemia)
 Alcohol hand gel
 Heel-raise foam pads
 List of WADA banned substances

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the clinician has access. A simple overview of player availability using a traffic-light coloured coding system (green = available, orange = available with conditions, red = unavailable) can help communication with coaching staff while maintaining confidentiality.

Working with a team provides an ideal opportunity to educate sportspeople and coaches. The medical team should look to create 'teaching moments' throughout the season during opportunities such as the pre-season health assessment, treatment sessions or team presentations. Experienced team clinicians have found that relevant topics of education include the following.

- **Sexual activity.** Team members should be warned of the dangers of acquiring sexually transmitted infections such as chlamydia, gonorrhoea, hepatitis B and HIV. Abstinence guarantees prevention, but condoms should be made available.
- **Hygiene.** Players should be encouraged to wash their hands regularly with alcohol hand gel and to avoid contact with any ill team members or loved ones. Travelling to foreign countries or participating in large sporting events places both athletes and support staff at risk of endemic disease and outbreaks of infectious diseases.¹⁴
- **Vaccinations.** Team members should be educated about the need for vaccinations depending on the countries they will compete in. Immunisations for the athlete are listed in Table 47.1. Team clinicians should use websites to gain up-to-date information on vaccination and illness prevention, especially in destinations where disease pattern is rapidly evolving.¹⁵
- **Anti-doping.** Team members must ensure they have a sound appreciation of those medications that are currently banned by the World Anti-Doping Agency (WADA). Up-to-date information can be obtained from national sporting organisations and Olympic

federations—some useful contacts are listed in Table 47.2. Medical team members need to educate players so they understand that many over-the-counter medications used in the treatment of coughs and colds are banned, and they should not take any medications or supplements without consulting their medical team.¹⁶

EMERGENCY ACTION PLANS

Emergency action plans (EAPs) are extremely useful documents that can help the sports clinician prepare for difficult scenarios that can arise within sports medicine. EAPs containing, but not limited to, the below information should be created for home training and competition venues in addition to all the locations you may travel to with your team:

- Contact details of the entire medical and coaching staff
- Address and contact details of all sporting and accommodation venues
- Distances from these locations to various specialist medical services and their contact details:
 - emergency department
 - trauma unit
 - orthopaedic surgery
 - plastic surgery
 - cardiac services
 - neurosurgery
 - burns unit
- National consulates and embassies are useful for foreign travel to source information on the quality of local medical personnel and facilities
- For each venue, entry points for emergency services (i.e. ambulances) should be determined so that clear instructions can be given to locate the team

Table 47.1 Immunisations suitable for the athlete according to the World Health Organization in May 2016

Routine vaccination	Selective use for travellers	Mandatory vaccination in particular countries
• Diphtheria, tetanus, pertussis • Hepatitis B • Haemophilus influenzae type b • Influenza (seasonal) • Measles, mumps, rubella • Pneumococcal disease • Human papillomavirus (females) • BCG	• Cholera • Hepatitis A • Japanese encephalitis • Meningococcal disease • Rabies • Tick-borne encephalitis • Typhoid • Dengue (CYD–TDV)	• Yellow fever (depending on the country) • Meningococcal disease and polio (required by Saudi Arabia for pilgrims; updates are available on www.who.int/wer)

Table 47.2 Contact details for obtaining drug information in various countries

Country	Drug information website address
World Anti-Doping Agency (WADA)	www.wada-ama.org www.wada-ama.org/en/Anti-Doping-Community/NADOs/List-of-NADOs/ lists all of the antidoping organisations in alphabetical order by country
Australia	Australian Sports Anti-Doping Authority, www.asada.gov.au
Brazil	Brazilian Agency for Doping Control, www.cob.org.br
Canada	Canadian Centre for Ethics in Sport, www.cces.ca
France	Agence française de lutte contre le dopage, www.afl.d.fr
India	National Anti-Doping Agency, India, www.nada.nic.in
Japan	Japan Anti-Doping Agency, www.playtruejapan.org
New Zealand	Drug Free Sport NZ, www.drugfreesport.org.nz
South Africa	SA Institute for Drug-Free Sport, www.drugfreesport.org.za
Spain	Agencia Estatal Antidopaje, www.aea.gob.es
United Kingdom	UK Sport, www.uksport.gov.uk
United States	US Anti-Doping Agency (USADA), www.usada.org

- All medical team staff should have completed an emergency life support training course that involves education regarding cervical spine immobilisation and transferring the athlete from the pitchside to hospital safely. The team should then make, and frequently practise, plans for caring for a medically unstable athlete in various situations
- A pitchside extrication plan should be created and practised regularly throughout the year. This plan should be exhaustive and address questions such as:
 - Who will perform what duties during pitchside extrication (such as cervical spine immobilisation, positions within a log roll, and positioning the spinal board)?
 - What will the instructions be during a log roll?
 - Does the defibrillator have scissors attached to it to facilitate quick removal of a jersey so pads can be applied to the chest?
 - Where are the defibrillator and other emergency equipment located is this equipment locked and if so who has the key?
 - Who will call for an ambulance?
 - How do you carry the athlete off the pitch?
 - Where will the athlete be taken to once on the spinal board?
 - Is this designated place warm or do blankets need to be stored there?
- Consider which member of the medical team will stay with an athlete during drug testing procedures or in the event that he or she is unable to travel back from a destination due to injury or illness.

PREPARING TO TRAVEL

When travelling to any venue away from home, the usual medical support structure is lost and the nature of existing medical support services in different countries varies substantially.¹⁷ In order to become self-sufficient, a detailed plan must be constructed before travelling with sports teams.¹⁸ Team clinicians may find it useful to split preparation into the following four stages;

1. Before travel

- *Travel destination.* The team clinician needs to have a good understanding of the climate, altitude, level of pollution, water supplies, security, recent health epidemics and natural disasters at the destination.¹⁹⁻²¹
- *Accommodation.* In hot climates, air conditioning may be advantageous for comfort although it may delay heat acclimatisation. It may also cause respiratory issues if used excessively as it can dry out the upper respiratory tract mucosa. Particularly tall athletes will require extra-long beds. Try to have a dedicated medical room when travelling with a large team to allow for privacy and confidentiality.
- *Vaccinations.* As discussed earlier (Table 47.1), vaccinations and malaria prophylaxis must be considered before embarking upon travel.¹⁵
- *Equipment and travel logistics.* It is important that the weight allowance for travel is considered when packing medical equipment so that everything can be transported to the destination. The local destination

medical services should be researched to decide whether medical equipment and medications can be sourced locally or need to be transported. It is important to note that medications may have different compositions in different countries. An advance visit to the competition location is extremely advantageous, if feasible, for the team physician to inspect local medical facilities, hotel kitchens and the like.²² Formulas exist within the literature to assist in estimating the pharmacological needs for a trip,²³ and it is important that all the medications be accompanied by a prescription or doctor's letter as well as being kept in the original containers.²⁴

- **EAPs.** As discussed above, creating EAPs to prepare for travelling with a team is invaluable.
- **Fitness to travel assessments.** Contact all team members, including coaches and officials, to ask about present and past injuries or illnesses. Often officials are older than the athletes and can cause anxiety for a travelling clinician because of medical conditions such as coronary artery disease. These individuals must be reminded to bring their own medications. Attempts must be made to treat injuries or illnesses prior to departure.
- **Nutrition.** It is challenging for athletes to maintain good dietary habits when travelling and they should be counselled on wise food choices before travel.²⁵ At large competitions, athletes often eat in village communal restaurants where buffet-style dining practices may lead to overeating. In addition to food intake, adequate hydration—especially when travelling to warm climates—is important.²¹ To minimise the risk of traveller's diarrhoea, athletes should eat only food that has been cooked, and avoid shellfish, salads, unpasteurised milk products and unpeeled fruit.
- **Jet lag.** This is discussed in the following section.
- **Insurance and indemnity.** As described earlier, your policies must cover you to practice at your destination. If any minors are in attendance, it is important to travel with the appropriate permission documents to treat them, as well as their guardian's contact information.²⁶
- **SAMPLE forms.** It is useful to create a history and treatment sheet for each athlete prior to travel in case she or he should need to be transferred to hospital during the trip. This sheet can be pre-filled with certain details such as their past medical history, regular medications, allergies, personal details and next of kin details. The SAMPLE format is a useful format for this sheet: **S**igns and symptoms, **A**llergies, regular **M**edications, **P**ast medical history, **L**ast meal/drink and **E**vents (including examination and treatment/intervention).

- **Injury prevention.** It is important to ensure that despite travelling, the team conduct proper warm-ups, stretching/yoga and strength maintenance, and receive regular soft tissue therapy.

2. During travel

Sports physicians should create a medical travel bag that contains medications to deal with travel sickness and simple analgesic medications. They must ensure that the athletes have sufficient water and food for the duration of their trip, especially if they are to compete soon after reaching their destination. If travelling long distances via road or aeroplane, the use of thromboembolism-deterrent (TED) stockings, electronic calf stimulators, earplugs, eye masks and facemask air humidifiers can help athletes to sleep in comfort and reduce the risk of venous thrombosis.

3. On arrival

A medical room should be established soon after arrival. Ideally, this should be a large room separate from any bedrooms. Treatment hours should be specified using an appointment sheet so that the practitioner has adequate time for meal and exercise breaks. This must be strictly enforced as athletes have a tendency to extend these hours. The clinician should ensure that athletes know the room number of the medical staff in case of emergencies. It is important that the team manager know how to contact the medical staff if the room is unattended. It also helps to obtain an athlete room list from the team manager.

For trips involving a single venue, a portable examination couch (treatment table) is invaluable for the comfort of both the athlete and the treating clinician. A low, soft bed may not be an appropriate site for massage or spinal mobilisation. On trips involving multiple venues, the advantage of having a treatment couch must be weighed up against the inconvenience of transporting it.

Traveller's diarrhoea is very common when travelling for sports,²⁷ and while the infection is usually self-limiting, team physicians should institute oral fluid replacement as the mainstay of treatment.

4. Journey home

Preparation for the return journey is important as some injuries or illnesses incurred while away may need specialist arrangements for travel or referral for specialist follow up.²⁸ Thought must be given to the thromboembolic risk of various injuries, such as fractures, and whether anticoagulation should be used prior to return travel. Once back in the home country, the medical staff should have a debriefing session to review any issues during the trip to enable changes to be implemented for future medical care.¹⁸

JET LAG

Jet lag presents a major challenge for clinicians travelling long distances with teams.

Air travel and jet lag

Air travel is an important part of professional and international sports. Short-distance air travel (up to 3 hours) does not appear to present any problems for the athlete.²⁹ However, extended air travel, often required for major events such as Olympic Games or World Championships, can provide significant problems.³⁰ Thus travelling athletes, team officials and team physicians will benefit from education on how to decrease the harmful effects of long-distance travel, such as jet lag, on performance (Table 47.3).

Jet lag is when the body is unable to adapt rapidly to a time zone shift and normal body rhythms lose synchrony with the environment. In general, it is a benign and self-limited condition. The major complaints of the condition are poor sleep, daytime fatigue and poor performance.³¹ The sleep deprivation secondary to the flight can be mostly made up in a day, whereas jet lag lasts much longer.³² A number of factors influence the severity of jet lag symptoms.³³⁻³⁴ These include: individual differences, number and direction of time zones crossed, temporal and seasonal timing of flight, age, impaired health, lack of previous travel experience, sleep deprivation, dehydration, stress, alcohol and excessive food intake.³⁵

Pathophysiology

The suprachiasmatic nuclei (SNC), located in the hypothalamus, produce a timing signal that modulates circadian rhythms of sleep and alertness, core body temperature and certain hormonal secretions such as melatonin and cortisol. The SNC, via the eyes, senses darkness and sends a timing signal to the pineal gland. This maintains the nocturnal secretion of melatonin for 10-12 hours. Beta blockers can inhibit melatonin secretion. Although the SNC signal is important, melatonin secretion requires darkness. Light intensity of >50 lux can cause some melatonin inhibition and light of >2000 lux completely suppresses it. Therefore, summer's long days and short nights will suppress melatonin secretion despite the 12-hour SNC 'on' signal.

The 12-hour SNC 'on' signal, despite melatonin inhibition by light, is considered an operational definition for biological night. Therefore, the nocturnal SNC signal synchronised to home night time (biological night) may result in melatonin secretion if the unadapted, jet-lagged traveller is placed in a dimly lit room. Melatonin can be measured in saliva, plasma and urine.³¹

The circadian rhythm is regularly synchronised to the 24-hour day by environmental time cues, termed 'zeitgebers', such as alternation of light and darkness, ingestion of melatonin, sleep/awake schedules, physical activity and meal timing.^{31, 35}

Travelling across multiple time zones (>3 time zones) causes a temporary misalignment between the circadian clock (lag) and the sleep/wake schedule at the destination time zone, that is slow to reset. The symptoms of jet lag dissipate as the circadian clock gradually resets (adjusts) to the time cues at the new destination time zone.^{31, 32, 35, 36} Although there is considerable individual variability, it is estimated that it takes about 1 day per time zone for the biological clock to resynchronise with the sleep/wake schedule.^{25, 31}

Prevention of jet lag

The faster the biological clock adapts to the new time zone, the shorter the symptomatic period. Thus, speeding up the adaptation is the primary goal.³⁵ When travelling in more than three time zones for a stay longer than five days, circadian adaptation is desirable (Table 47.3).^{35, 37} Pre-flight adjustment to travel may speed up adaptation and is discussed below.³⁸ Experienced practitioners and many successful athletes also suggest that low-intensity physical activity early after arrival helps promote adjustment to the new time zone. Seasoned air travellers have developed a series of guidelines to minimise the adverse effects of long-distance travel (Tables 47.3 and 47.4).

Timed light exposure and avoidance

Light intensity and timing are the two most important factors that influence phase shift. Exposure to bright light (sunlight at approximately 3000-100 000 lux, light box at 1500-3500 lux or room light at 100-550 lux) can help phase shifting. The magnitude of shift is greater as we get closer to the Tmin (the body's minimum core temperature, which is estimated to occur at 3-5 am in most individuals). Light has more shifting effect at night when bright light is absent.

PRACTICE PEARL

Light exposure late in the sleep episode or in the early morning (i.e. after the Tmin phase advances) will promote phase shifting. The same light stimulus, when applied early in the sleep episode (i.e. before the Tmin) will delay phase shifting.^{31, 32, 35, 39}

Table 47.3 General guidelines for adaptation strategies based on the length of stay at destination and numbers of zones crossed³⁵

Length of stay at destination	Recommended strategy
Long (>5 days)	• Pre-travel adaptation of sleep schedule to that of the destination • Timed light exposure and avoidance • Timed melatonin intake
Intermediate (3–4 days) OR Short (1–2 days) OR Game aligned with circadian time of peak performance	• Work around sleep and alertness times • Schedule critical activity to daytime in the departure zone • Airplane light exposure/avoidance strategy • Short-term measure to maintain alertness (caffeine) and sleep (naps and short-acting hypnotics)

Table 47.4 General principles to help clinicians customise jet travel schedules for athletes travelling in any direction across multiple time zones

Goal: Get Tmin within sleeping episodes and avoid being awake at the sleepest time (Tmin)

Eastward jet lag	Westward jet lag
Differences	
• Requires advancing the circadian clock • Harder and slower to adjust (1 h/day) • Abrupt 12-h time zone shift, takes 8–9 days to adjust • Destination Tmin = origin Tmin + number of time zones crossed	• Requires delaying the circadian clock • Easier and quicker to adjust (1.5–2 h/day) • Abrupt 12 h time zone shift, takes 4–5 days to adjust • Destination Tmin = origin Tmin – number of time zones crossed
Similarities	
• Maximum or largest shift is 12 h time zones • Traveller feels sleepy in hours surrounding Tmin and difficulty sleeping at hours far from the Tmin • Traveller will experience impaired daytime activity near Tmin • Phase shifting needs to be applied daily for several days to produce results	

Tmin origin = minimum core temperature, estimated to occur at 3–5 am (hours before wake-up time) for most individuals, 2–4 am for larks (morning type, M-type), and 5–7 am for owls (evening type, E-type) based on the sleep pattern for the last week. Tmin coincides roughly with the maximum nocturnal melatonin level in the circulation

Timed light avoidance is equally important. Light avoidance can be achieved by staying in a dim light room (<10 lux) or wearing dark sunglasses.³⁹ Light exposure should be augmented with light avoidance during flight or on arrival to a new destination as it ensures a unidirectional phase shifting. For instance, an eastward-travelling athlete needs to maximise light exposure after Tmin and avoid or minimise light before Tmin to ensure phase advancement of the circadian clock, and vice versa for the westward travelling athlete.³⁸ On the other hand, exposure to light before and after Tmin would result in no phase shifting.

Timed melatonin pills

Despite being marketed as an over-the-counter sleep aid, melatonin is not associated with sleep or increased sleepiness. Instead, melatonin is considered a darkness signal. The timing of melatonin pills is more important than dose. It has more of a phase-shifting effect when there is less endogenous melatonin in the circulation. Therefore, a timed melatonin pill (3 mg) can help produce the greatest phase advance of the circadian clock when taken in the afternoon (approximately 5 hours before dim light melatonin onset [DLMO] or 12 hours before Tmin) while phase delays occur when taken in the morning.^{32, 35}

Pre-travel sleeping schedule

Adjusting sleep schedule in 1-2 days (depending on the number of time zones to be crossed) prior to travel to gradually match the destination schedule may help with phase shifting. Sleep on flights should be avoided unless it is night at destination.⁴⁰

Synergistic approach

When combined with a pre-travel sleep schedule for the destination, timed light exposure/avoidance and melatonin pills can have a synergistic phase-shifting effect.

Symptomatic treatment for jet lag

If the travelling athlete still suffers jet lag symptoms despite the gradual shifting, stimulants such as slow-release oral caffeine (300 mg) could be helpful to induce daytime alertness, and hypnotics such as oral zolpidem (10 mg) could be used to counteract night insomnia.

REFERENCES

References for this chapter can be found at www.mhhe.com/au/CSM5e

